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What you will learn in this chapter

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What you will learn in this chapter:

- a criterion to make **optimal hypotheses** in learning tasks,
- a method to solve **analogies**,
- a new way to **detect anomalies**,
- a new understanding of machine learning as a way to achieve compression,
- a method to define an unbeatable AI.

Detailed outline Chapter (4)

Machine Learning and Algorithmic Information

The induction problem

Induction is one of the oldest philosophical problem. And Algorithmic Information Theory solved it! As an example, we will show how Kolmogorov complexity can be used to make *analogies*.

Minimum Description Length

The most common way to perform induction in practice is called MDL: *Minimum Description Length*.

Prototype-based models and compression

The central topic of this chapter is that *machine learning* translates into *compression*. This claim will be illustrated with *prototype-based models*.

Conversely, compression provides a criterion that can be used to guide machine learning. It is illustrated in the context of *clustering*.

Anomaly detection

Anomaly detection is an important and difficult topic of AI. Anomalies will be shown to be events that are abnormally *simple*.

Universal induction

Are you dreaming of an AI system that would beat us all? AIT offers precisely this, at least in principle.

- First as an ideal way to perform *prediction*, using what is called the *universal distribution*.
- Then as an ideal way to perform action: we will briefly consider *AIXI*, an ideal version of reinforcement learning based on algorithmic information.



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